Toward a four-dimensional Yang–Mills construction

Notebook 30: quantum descent and renormalized block stability

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Abstract

We continue the existence argument by constructing the local quantum update on the complete interaction quotient of Notebook 29. Actual finite minimum comparisons supply connecting gauges that, in the infinite local limit, approach one constant exponentially. Compact gauge approximation then proves that classical substitution preserves local gauge invariance for every invariant input class. A norm-preserving periodic construction identifies finite normalized interactions independently of their carrier representation, including wrapping supports and exact finite Ward identities.

The actual strict Gaussian expansion is continuous in its local analytic data. Its proof uses support incidence, strict marked-order tails, an explicit positive inherited-cardinality margin and finite dimensional dominated convergence. These estimates define the homogeneous local quantum map on the invariant quotient and bound the input dependence of its quantum remainder by $C(gT/b)$. Finite translation averaging identifies limits of varying finite inputs with the same map, giving compatibility for every fixed finite history inside the common budgets.

We then examine the renormalized recurrence using the actual classical reference actions and the finite endpoint ratio R^q. Exact reference transport cancels intermediate classical projections. A fixed outer field domain and vanishing total linear jet give an $O(b^2)$ zero-input quantum force on the inner radius b. The extracted activities therefore have an $O(b^2)$ invariant envelope under a finite block of sufficiently weak steps, with all intermediate inputs in the common domain. Every quantum correction and vacuum factor remains present. These estimates concern the homogeneous local sector relative to one fixed joint classical system. The running coupling must obey its actual extraction equation; a prescribed sequence of small parameters does not by itself furnish that trajectory. Mixed and large-field histories, sourced bounds and the continuum observable theory remain necessary for existence.

The objective is a nontrivial four-dimensional Yang–Mills quantum theory. All results here use the finite constructions of Notebooks 15–29 with their stated common domains and actual retained-link windows. The proofs below add explicit bridges between finite normalized integrals, infinite analytic interaction classes and renormalized activity estimates. They do not assume that equality of infinite compact-field totals automatically identifies a finite integral, nor that a final norm bound alone proves continuity.

1 Summable connecting gauges and the invariant quotient

This section closes the gauge-invariance interface for classical substitution on the analytic interaction quotient. Its finite-regulator inputs are the actual ordinary-minimum comparison and equality theorem, the ordinary-to-corrected transfer, the constraint-preserving flat-slice flow, and the coherent pre-gauge support estimates. It does not assume a weighted estimate for the nonlinear gauge-fixing inverse.

1.1 Joint limits, domains and the statement

Fix a preceding homogeneous refinement M, a finite allowed endpoint ratio L, and a single joint subsequence of compatible cubic regulators. Let

$$e^{ia_{L,N}(w)} = P_{M,L,N}(e^{iw}) \longrightarrow e^{ia_L(w)} \tag{1}$$

locally at every current link, locally uniformly in the finitely many variables of each compactly supported final field w. The current lattice has spacing $1/L$ in final physical units. Assume the established coherent support bound

$$\|a_L(w)|_i\|_\infty \le \rho e^{-\kappa \operatorname{dist}(i,K)}, \qquad \rho = \frac{C_A b}{L}, \qquad \|w\|_\infty < b, \quad \operatorname{supp} w \subset K. \tag{2}$$

Here K is a finite set of final cells. The same assertion holds for every other compact datum in the ball. Joint analytic compactness and the finite-regulator support estimates supply this input; unrelated pointwise choices of subsequences do not replace it.

Use a unitarily invariant matrix norm for group elements. Its comparison with the scalar-coordinate maximum norm defining the activity polydisc has fixed color-dependent constants. In particular, constant conjugation is not silently treated as an isometry of that coordinate maximum norm. All such comparison constants are included below. The dimension is $d \ge 2$, with $d = 4$ in the application. Constants in summability estimates may depend on fixed L and on a finite compact support, but not on preceding refinement or approximating volume.

Let $\mathcal{Q}_{\mathrm{inv}}$ be the quotient of anchored analytic interactions by equality of their vacuum-normalized totals on compact fields. Its invariant totals are invariant under constant global adjoint transformations and under compact local gauge paths that stay inside the analytic field ball. An identity for arbitrary disconnected endpoint gauges is not part of the conclusion. The summable-field extension of the quotient gives, for $0 < \rho_1 < b$,

$$|\Phi_E(A) - \Phi_E(B)| \le \frac{\|[E]\|_{\mathcal{Q}}}{b - \rho_1} \|A - B\|_{\ell^1_{\mathrm{cell}}}, \qquad \|A\|_\infty, \|B\|_\infty \le \rho_1. \tag{3}$$

The normalized total is absolutely convergent on these summable fields, and its value depends only on the class.

Theorem 1.1 (Universal local gauge identity for a fixed joint map)**.** *Under these inputs, the scalar substitution*

$$[E] \longmapsto (w \longmapsto \Phi_E(a_L(w))) \tag{4}$$

preserves local gauge invariance for every invariant input class on a common smaller germ. If

$$C_g C_A / L < 1/2, \tag{5}$$

where C_g is the fixed group/chart constant constructed below, the compact connecting paths have a common strict margin in the original radius-b field ball. The local analytic identity then holds throughout the connected common coarse domain:

$$\Phi_E(a_L(w^v)) = \Phi_E(a_L(w)) \tag{6}$$

for the allowed compact local gauge transformations.

The connecting gauges and domain choices are independent of E. Consequently, when the separately established connected representation and boundedness estimates for classical substitution apply, (4) is an action on the whole invariant quotient for this fixed joint background system.

1.2 The small gauge supplied by the actual equality proof

We record the quantitative consequence of the ordinary-minimum equality proof. For an ordinary-feasible fine field $U = e^{i\eta A}$, the first transfer uses $g_1 = e^{iH_b\theta}$. Its induced-gauge implicit equation and the cumulative gauge $b_j(U)$ obey

$$\|\theta\|_\infty \le 2\|\omega\|_\infty, \qquad \omega = \log b_j(U)/i, \qquad \|\omega\|_\infty \le C\|A\|_{C^{1,\alpha}}. \tag{7}$$

The proved bound $H_b : \ell^\infty \to C^{2,\alpha}$ controls this gauge uniformly in mesh and volume. The constraint-preserving flat-slice flow has scalar velocity

$$\lambda(t) = \mathcal{L}_{A(t)}A(t), \qquad \|\lambda(t)\|_{C^{2,\alpha}} \le Ce^{-t}\|R_0 D^T A(0)\|_{C^\alpha} \le Ce^{-t}\|A(0)\|_{C^{1,\alpha}}. \tag{8}$$

Integrating its actual unitary gauge ODE therefore gives

$$\|g_2 - I\|_\infty \le C\int_0^\infty \|\lambda(t)\|_\infty \, dt \le C\|A(0)\|_{C^{1,\alpha}}. \tag{9}$$

Finally the selected ordinary representative is obtained from the strict corrected minimizer by $h_* = e^{iE_{\mathrm{pt}}\omega_*}$, where $\|\omega_*\|_\infty \le C\|B\|_\infty$ and $E_{\mathrm{pt}} : \ell^\infty \to C^{2,\alpha}$ is the fixed smooth point-value extension.

If equality holds in the two minimum comparisons, strict convexity on the fixed slice identifies the intervening field with that corrected minimizer. The product of the actual transfer gauges consequently maps U to the selected ordinary minimizer and satisfies

$$\|g_{\mathrm{eq}} - I\|_\infty \le C(\|A\|_{C^{1,\alpha}} + \|B\|_\infty). \tag{10}$$

The constants are uniform on the established smaller comparison balls. This is stronger than equality of action values alone. It uses no inverse massless gauge Laplacian and no unproved spatial locality of $\mathcal{L}_A$.

For a compact coarse gauge $v = e^{i\theta}$, extend its prescribed values to the original fine lattice by $\widehat{v} = e^{iE_{\mathrm{pt}}\theta}$. This extension is compact in physical space and has $C^{2,\alpha}$ norm bounded by $C\|\theta\|_\infty$, uniformly in the fine mesh. The fields $U_{\mathrm{ord}}(w)^{\widehat{v}}$ and $U_{\mathrm{ord}}(w^v)$ obey the same ordinary constraint. The two minimum comparisons, one using the inverse gauge, give equality. Equation (10) therefore provides an actual fine gauge g_N with

$$U_{\mathrm{ord,N}}(w^v) = U_{\mathrm{ord,N}}(w)^{g_N}, \qquad \|g_N - I\|_\infty \le C(b + \|\theta\|_\infty) \le \delta. \tag{11}$$

Choose the data and local gauge neighborhoods so that δ is a fixed small number. Every competitor has complete fine-potential norm at most $C(b + \|\theta\|_\infty)$ and hence lies in the actual comparison class. Ordinary covariance of the prefix gives

$$e^{ia_{L,N}(w^v)} = \big(e^{ia_{L,N}(w)}\big)^{g_N|_{\mathrm{current}}}. \tag{12}$$

1.3 An infinite connecting gauge and its tail

Fix real compact w, θ in these neighborhoods. The endpoint configurations already converge along the specified joint subsequence. Compactness of the group and a diagonal subsequence over current sites give a limiting gauge for this pair. Taking a further subsequence of gauges changes neither specified endpoint limit. With $U = e^{ia_L(w)}$ and $V = e^{ia_L(w^v)}$, local passage gives

$$V_{xy} = g(x)U_{xy}g(y)^{-1}, \qquad \sup_x \|g(x) - I\| \le \delta. \tag{13}$$

No simultaneous or holomorphic choice of g for all pairs is required. Both endpoints have exponential decay because w, w^v are compact.

Lemma 1.2 (One constant at infinity)**.** *Let compact-group link fields U, V on $\mathbb{Z}^d$, $d \geq 2$, satisfy $V = U^g$. Suppose their positive bonds based at x obey*

$$\|U_b - I\| + \|V_b - I\| \leq C_K e^{-\mu|x|_\infty}. \tag{14}$$

Then there is a single $c \in G$ such that

$$\|g(x) - c\| \leq C'_K(1 + |x|_\infty)e^{-\mu|x|_\infty}. \tag{15}$$

Thus $c^{-1}g - I$ is summable. If $\|g - I\|_\infty \leq \delta$, its supremum is at most 2δ.

Proof. The edge equation implies $g(y) = V_{xy}^{-1} g(x) U_{xy}$. Unitary invariance of the matrix norm gives

$$\|g(y) - g(x)\| \leq \|U_{xy} - I\| + \|V_{xy} - I\|. \tag{16}$$

The ray $g(ne_1)$ is Cauchy, with tail $O(e^{-\mu n})$, and its limit c belongs to G. For $n = |x|_\infty$, the lattice boundary of $[-n, n]^d$ connects x to ne_1 by a path of length at most $C_d n$. A boundary coordinate can be held at $\pm n$ while the other coordinates are adjusted; a second boundary coordinate permits changing its sign. This uses $d \geq 2$. Each positive lower endpoint has norm at least $n - 1$. Summing (16) along the path gives $C_d n C_K e^{-\mu(n-1)}$. Combining this with the ray tail proves (15). Polynomial shell growth proves summability. Taking the limit in the small-supremum bound and using $\|c^{-1}g - I\| = \|g - c\|$ gives the last assertion. □

For the actual fields, (2) implies (14) with μ a positive multiple of κ/L, after enclosing their finite final supports in a cube. Support-dependent constants are harmless for equality of evaluations. In one dimension the two ends can have different constants; the restriction $d \geq 2$ is substantive.

1.4 Compact gauge paths with a common strict margin

Put $h = c^{-1}g$. Then $h - I \in \ell^1$ and

$$U^h = c^{-1} V c. \tag{17}$$

For sufficiently small δ choose the common Hermitian logarithm $h(x) = e^{i\lambda(x)}$. It has small supremum and belongs to ℓ^1. Let λ_n equal λ inside a current cube of radius n and vanish outside, and set $h_n = e^{i\lambda_n}$. These gauges are compact and

$$\|h_n - h\|_{\ell^1} \longrightarrow 0, \qquad \|U_b^{h_n} - U_b^h\| \leq \|h_n(x) - h(x)\| + \|h_n(y) - h(y)\|. \tag{18}$$

Hence the transformed links, and their potentials on a common strict logarithm chart, converge in cell ℓ^1. A smooth physical cutoff is not needed on this fixed current lattice: the truncation jump itself decays exponentially.

The paths, not just their endpoints, stay in the analytic domain. The edge equation and the Lipschitz bound for the small logarithm give

$$\|\lambda(y) - \lambda(x)\| \leq C(\|a_L(w)_b\| + \|a_L(w^v)_b\|) \leq C\rho. \tag{19}$$

For all sufficiently large n, the exterior truncation jump of λ_n is also at most ρ. Duhamel's formula for real Hermitian matrices yields, for $0 \leq t \leq 1$,

$$\|e^{it\lambda_n(y)} - e^{it\lambda_n(x)}\| \leq \|\lambda_n(y) - \lambda_n(x)\|.$$

The small link logarithm and fixed color-norm comparisons consequently give

$$\sup_{0 \leq t \leq 1} \left\| \frac{1}{i} \log(U^{e^{it\lambda_n}}) \right\|_\infty \leq C_g \rho. \tag{20}$$

Here C_g is independent of support, truncation radius, refinement and volume. Condition (5) places these paths strictly below $b/2$. At a smaller fixed endpoint ratio, restricting the data radius gives a local germ instead; a whole-ball margin is not inferred without the condition.

The input field in (20) is summable rather than compact. Before invoking the defining invariance of the quotient, truncate its potential to a much larger cube containing the support of λ_n and all adjacent bonds. Inside the gauge support this compact input is unchanged. Outside, the gauge is the identity and the field is either unchanged or zero. The same strict path bound therefore holds. Compact-field local gauge invariance and then (3), as the input truncation tends to infinity, give

$$\Phi_E(\log U^{h_n}/i) = \Phi_E(\log U/i). \tag{21}$$

Now send n to infinity using (18) and (3). Extend constant global adjoint invariance to summable fields by the same compact-input approximation. The result is

$$\Phi_E(a_L(w)) = \Phi_E(\mathrm{Ad}_{c^{-1}}\, a_L(w^v)) = \Phi_E(a_L(w^v)). \tag{22}$$

This proof uses the same gauges and margins for every invariant class.

1.5 Analytic continuation and precise scope

To finish Theorem 1.1, differentiate (22) at zero compact coarse gauge parameter. The resulting infinitesimal identity is holomorphic in the finitely many coarse field variables on their connected product ball and vanishes on a real open set. It therefore vanishes throughout that complex ball. Integrating it along any allowed compact local gauge path that stays in the ball proves (6). This argument does not extend an infinitesimal Ward identity to arbitrary disconnected endpoint transformations.

The same endpoint method preserves the spatial symmetries required by the invariant quotient, when the actual finite blocking and minimum problem have those symmetries. The transformed ordinary minimizer and the selected minimizer at transformed data are small feasible fields with equal minimum values. The actual equality proof supplies the controlled gauge; prefix covariance gives the endpoint orbit relation; compact data give exponential tails. The preceding compact-gauge argument applies, and the input total's spatial symmetry supplies the remaining change of variables. Constant adjoint covariance is exact for the equivariant construction, or follows from the small comparisons and connectedness of the compact group. Symmetry of individual carrier coefficients is unnecessary.

Universality means that every admissible interaction class can be evaluated using this one fixed joint infinite background system. It does not assert uniqueness among different regulator subsequences. Semigroup identities require the maps and their compositions to belong to the same joint system; finite-regulator identities alone do not establish consistency for unrelated limits.

The connected representation and boundedness of the quotient action remain separate analytic inputs. This section proves their missing universal gauge identity from the actual finite transfer and flow, endpoint decay, and summable-field continuity. It concerns homogeneous small fields, $d \geq 2$, and compact local gauge paths. It proves neither quantum-integral descent on the quotient nor quantum induction, large-orbit or mixed-history control, or a continuum measure.

2 Bounded finite periodization of analytic interaction classes

Use the infinite activity space and normalized-total quotient of Notebook29. Fix the cell size J, at most D scalar coordinates per cell, exponent $a > 0$, and radius $b > 0$. Carriers are nonempty finite connected subsets of the cell graph $\mathbb{Z}^d$, and

$$\|E\|_{a,b} = \sup_i \sum_{H \ni i} e^{a|H|} \|E_H\|_b, \qquad \Phi_E(A) = \sum_H [E_H(A|_H) - E_H(0)] \tag{23}$$

on compact fields. Let $\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^d$; the underlying current lattice has period NJ. Write π_N for cell and coordinate projection and A^{per} for the infinite periodic lift of a finite field. It is bounded, generally not summable. Internal link directions and color types are identified by their actual periodic conventions.

Theorem 2.1 (Canonical normalized periodization)**.** *Every class with a J-cell-translation-invariant normalized total has a translation-covariant representative without increasing the norm of any chosen input representative. There is a canonical linear periodization map*

$$\mathrm{Per}_N : \mathcal{Q}^{\mathrm{tr}}_{a,b} \longrightarrow \mathcal{Q}^N_{a,b}, \qquad \| \mathrm{Per}_N \| \le 1, \tag{24}$$

where the finite quotient uses connected torus carriers and equality of vacuum-normalized totals. Its scalar functional is holomorphic on the original open radius-b polydisc. Every fixed derivative tensor is the exact relative periodization of the infinite total tensor. Total spatial and color symmetries pass whenever they preserve the torus. Local gauge invariance passes on the specified common logarithmic chart.

The proof retains raw constants in carrier norms and normalizes only the total. It does not identify the result with an unrelated actual finite-regulator quantum output.

2.1 Translation averaging without norm loss

Let τ_t translate carriers, their coordinate slots and their functions by a cell-lattice vector t. It is an isometry of the activity space. If $[E]$ is translation invariant, then $[\tau_t E] = [E]$. For centered Følner cubes F_m, put

$$E^{(m)} = \frac{1}{|F_m|} \sum_{t \in F_m} \tau_t E. \tag{25}$$

Its norm is at most $\|E\|_{a,b}$, and for every fixed s,

$$\|\tau_s E^{(m)} - E^{(m)}\|_{a,b} \le \frac{|(F_m + s) \triangle F_m|}{|F_m|} \|E\|_{a,b} \longrightarrow 0. \tag{26}$$

Each fixed carrier has finitely many input coordinates. Countable Montel extraction gives locally uniform limits E^{cov}_H for all carriers. Fatou, first on finite carrier collections and smaller polydiscs and then at the original open radius, yields

$$\|E^{\mathrm{cov}}\|_{a,b} \le \|E\|_{a,b}. \tag{27}$$

Equation (26) passes componentwise and gives exact covariance. On a compact input supported in K, the omitted carriers with $|H| > Q$ contribute at most $2|K|\|E\|_{a,b} e^{-aQ}$. The finite remaining sum converges, so $\Phi_{E^{\mathrm{cov}}} = \Phi_E$. No uniqueness of this covariant representative is needed.

Only cell translations are averaged. Finer unit translations, point symmetries and color invariance need not hold for the individual representative and will pass through its total.

2.2 The exact orbit count and wrapped carrier norm

For a covariant representative define

$$U^E_N(A) = \sum_{i \in \Lambda_N} \sum_{H \ni i} \frac{E_H(A^{\mathrm{per}}|_H)}{|H|}, \qquad \Phi^{[E]}_N(A) = U^E_N(A) - U^E_N(0). \tag{28}$$

Choose one fundamental representative for each cell i. The formula is independent of that choice by covariance. Its series is absolutely convergent in the supremum norm on the full open b polydisc, bounded by $N^d\|E\|_{a,b}$ before normalization.

Let $\mathcal{O}$ be an orbit of finite carriers under $N\mathbb{Z}^d$. A nonempty finite carrier has no nonzero translation stabilizer. Every point in a representative H corresponds to exactly one incidence pair (i, H') with $i \in \Lambda_N$, $H' \in \mathcal{O}$ and $i \in H'$. Hence

$$\sum_{i\in\Lambda_N} \sum_{\substack{H'\in\mathcal{O}\\ i\in H'}} \frac{1}{|H'|} = 1. \tag{29}$$

Covariance and periodicity make the summand value constant on the orbit. Therefore

$$U_N^E(A) = \sum_{\mathcal{O}} E_H(A^{\mathrm{per}}|_H), \qquad H \text{ any representative of } \mathcal{O}. \tag{30}$$

This counts each carrier orbit once, including those whose projection is not injective.

For a connected nonempty torus carrier K, set

$$E_{N,K}(A|_K) = \sum_{\mathcal{O}:\,\pi_N H = K} E_H(A^{\mathrm{per}}|_H). \tag{31}$$

Coordinate repetition is a linear map of maximum norm one. Every summand is holomorphic on the original b polydisc and strictly supported on K. Projection preserves connectedness.

Fix a torus cell j and one infinite lift. For every orbit meeting j, choose one representative containing that lift. Distinct orbits give distinct carriers. Since $|\pi_N H| \le |H|$, this gives

$$\begin{aligned}\sum_{K\ni j} e^{a|K|}\|E_{N,K}\|_b &\le \sum_{\mathcal{O}:\, j\in\pi_N H} e^{a|\pi_N H|}\|E_H\|_b \\ &\le \sum_{H\ni j} e^{a|H|}\|E_H\|_b \le \|E\|_{a,b}.\end{aligned} \tag{32}$$

This proves normal convergence of the grouped family and the no-loss carrier estimate. Choosing one incidence here is an upper bound for the norm; the exact energy multiplicities were already counted in (29). No factor $|\pi_N H|/|H|$ belongs in the energy.

The raw grouped coefficients retain their constants. Normalizing each coefficient separately could cost a factor two, and is unnecessary. For example the constant singleton family $E_{\{i\}} = c$ is the zero normalized class, but its raw periodic total is $N^d c$. A canonical unnormalized exponential therefore needs an additional vacuum convention.

2.3 Derivatives, representative independence and local stability

Let $K_n^E(i_1, \alpha_1; \ldots; i_n, \alpha_n)$ be the infinite total derivative kernel, with cell indices and internal types separated. At every fixed n, connected-carrier Cauchy estimates give absolute anchored summability with a positive spatial exponent. Constants can depend on n, D, J, a, b; no uniform all-order absolute Taylor norm is being asserted.

Differentiate (30). A derivative in a periodic coordinate differentiates each old occurrence of that coordinate. Translate its first occurrence $i_1 + Nz_1$ to a fixed lift i_1. Every carrier through i_1 occurs exactly once. The remaining slots then range over all their periodic lifts. Thus

$$\begin{aligned}&D^n U_N^E(0)[e_{i_1,\alpha_1}, \ldots, e_{i_n,\alpha_n}] \\ &\quad = \sum_{z_2,\ldots,z_n\in\mathbb{Z}^d} K_n^E(i_1, \alpha_1; i_2 + Nz_2, \alpha_2; \ldots; i_n + Nz_n, \alpha_n).\end{aligned} \tag{33}$$

Repeated finite slots include both coincident and distinct infinite lifts with their ordinary multilinear multiplicities. Absolute anchored summability justifies the rearrangement.

A null class has all $K_n = 0$ for $n \geq 1$. Its normalized periodic total has every Taylor coefficient zero and hence is zero on the connected finite-dimensional b polydisc. Two covariant representatives of the same class therefore give the same normalized energy. This proves independence of the translation-averaging subsequence. The scalar map is linear even if representative selection is not. Taking representatives approaching the quotient infimum in (27),(32) proves (24), and

$$|\Phi_N^{[E]}(A)| \leq 2N^d \|[E]\|_{\mathcal{Q}}. \tag{34}$$

There is exact scalar tiling consistency. For an integer $q > 0$, repeat A on Λ_{qN}. The density in (28) is then N-periodic, so

$$\Phi_{qN}^{[E]}(A \text{ repeated}) = q^d \Phi_N^{[E]}(A). \tag{35}$$

A rectangular multiple has its corresponding volume factor.

There is also eventual exact stabilization of each fixed lifted coefficient. Let K be a fixed finite connected infinite carrier and choose, conservatively, $N > 4\,\mathrm{diam}(K) + 4$. The copies $K + Nz$ are disjoint and nonadjacent. A connected H with $\pi_N H = \pi_N K$ lies in their union, hence in one copy; surjectivity of its projection makes it equal to that copy. Equation (31) consequently has one orbit, and under lifted coordinates

$$E_{N,\pi_N K} = E_K \tag{36}$$

on the full original polydisc. This gives local normal convergence of canonical periodizations without a winding-tail estimate for fixed K. It asserts neither global norm convergence nor stabilization on carriers growing with N.

2.4 Local gradients and exact finite gauge invariance

For a fixed infinite scalar coordinate i and any bounded field $\|A\|_\infty \leq \rho < b$, define

$$G_i^E(A) = \sum_{H \ni \mathrm{cell}(i)} \partial_i E_H(A|_H). \tag{37}$$

Directional Cauchy gives

$$|G_i^E(A)| \leq \frac{\|E\|_{a,b}}{b - \rho}. \tag{38}$$

If A, B agree in a Q-cell neighborhood of i, each differing summand has a connected carrier of size at least cQ. Thus

$$|G_i^E(A) - G_i^E(B)| \leq \frac{2\|E\|_{a,b}}{b - \rho} e^{-caQ}. \tag{39}$$

This proves continuity under compact truncation of bounded fields. On compact fields G_i^E is the derivative of Φ_E. A null family therefore has $G_i^E = 0$ on all bounded interior fields, even without covariance of that representative.

For a covariant representative, the same first-slot reanchoring used above works at any periodic interior field and gives

$$\partial_i U_N^E(A) = G_i^E(A^{\mathrm{per}}). \tag{40}$$

This is another proof that null classes periodize to constant raw totals and to zero normalized totals.

Suppose the infinite normalized total is locally gauge invariant. For a fixed unit-current site v, its infinitesimal logarithmic gauge field $\mathcal{L}_v(A)$ reads only the incident bond potentials and is supported on those finitely many bonds. On compact fields the Ward identity is

$$\sum_{i \text{ incident to } v} G_i^E(A) \mathcal{L}_{v,i}(A) = 0. \tag{41}$$

Compactly truncate an arbitrary bounded interior A. The local generator values stabilize, and (39) passes the gradient terms. Thus (41) holds for bounded periodic fields too. Using (40), identify all incident orientations and periodic multiplicities. The finite total obeys

$$D\Phi_N^{[E]}(A)[\mathcal{L}_{\lambda,N}(A)] = 0 \tag{42}$$

for every finite torus gauge parameter. No noncompact infinite gauge path has been applied to Φ_E; only a local gradient identity was extended by absolute convergence.

The logarithmic generator must be holomorphic on the domain. If b lies inside its fixed chart radius, the assertion holds on the whole open b polydisc. Otherwise keep full-b holomorphy of the interaction and restrict the gauge assertion explicitly to $b_g = \min(b/2, r_{\log}/2)$, where $r_{\log}$ is a fixed radius of holomorphy for that generator. The actual small-field setup already chooses b inside this chart. Finite gauge invariance follows along any gauge path whose fields stay in the common domain. Large paths leaving it are not covered.

All finite nonlinear Ward equations also pass coefficientwise in field degree. At any fixed order, differentiating the compact infinite identity uses finitely many local generator coefficients, nearest-neighbor shifts and coincident-slot embeddings of total kernels. Equation (33) commutes with these operations: summing images turns an infinite coincidence condition into the exact periodic one. Fixed-order summability justifies it. Equivalently, differentiate the exact analytic identity (42).

2.5 Other symmetries and the finite-history boundary

Any finer translation, hypercubic or global color symmetry of the infinite total is inherited by its kernels. Relative periodization commutes with each symmetry preserving the torus. Taylor uniqueness gives the same symmetry of the normalized finite total. Negative bonds use the signed reversal of their positive-bond logarithmic coordinate. No symmetry of individual $E_{N,K}$ is needed. Reality follows similarly; a real covariant representative can also be obtained by averaging with its complex-conjugate reflection without norm increase.

This completes Theorem 2.1. The canonical finite functionals have exact Ward identities and a no-loss connected representation. They need not equal an unrelated actual finite-regulator family. Two uniformly bounded finite sequences can have the same infinite local interaction while differing by winding-supported terms at every finite size. Compactness does not assert those terms are exactly absent.

Consequently, inserting $\mathrm{Per}_N[E]$ into a quantum integral requires either an actual history/normalization comparison or a construction using these periodized inputs from the outset. Stationary backgrounds, retained windows, Gaussian normalizations and gauge factors are not identified by periodizing the interaction alone. Raw exponentials also require a separately specified vacuum convention. No quantum integration identity, continuity of the actual quantum update, or infinite induction is proved here.

3 A local limit and quotient descent of the quantum macrostep

Fix the composite ratio R, auxiliary grouping J, preceding homogeneous depth M, real retained link/color windows, cutoff threshold T, positive coupling g, and common field radius b from the common-domain construction. All fluctuation-cell dimensions are fixed once R, J are fixed. Compatible cubic tori exhaust the current and next lattices together. Write

$$\|E\|_{a_0,b} = \sup_i \sum_{H \ni i} e^{a_0|H|} \|E_H\|_b, \qquad a_0 = 1/16,$$

for the connected activity norm. Quotient classes identify vacuum-normalized totals on compact fields, as in Notebook29.

The actual finite local correction has the decomposition

$$\mathcal{F}_N(E_N) = \mathcal{T}_N E_N + \Delta_N + \mathcal{Q}_{\mathrm{det},N} + \mathcal{L}_N(Q_{E_N}),$$
$$\mathcal{G}_N(E_N) = \Delta_N + \mathcal{Q}_{\mathrm{det},N} + \mathcal{L}_N(Q_{E_N}). \tag{43}$$

Here $\mathcal{T}_N$ is pre-gauge background substitution, Q_{E_N} is the inherited fluctuation difference in the actual chart, and $\mathcal{L}_N$ is the strict Gaussian cluster logarithm with its fixed classical base factors and precision. The pivot term, determinant, sharp windows and field-independent normalization are those of the finite common-domain integral. Only after constructing this expression do we subtract its value at zero background to form a quotient total. A separately tracked vacuum normalization is not identified with zero.

3.1 One fixed, input-independent classical system

Call coefficients *locally normal* when every fixed lifted finite carrier coefficient converges uniformly on compact subsets of its finite-dimensional background polydisc. For joint coefficients require this convergence uniformly on the fixed real window of their finitely many fluctuation coordinates. Matrix coefficients use their finite block norm. The final norm bound is on the original open b ball; local normal convergence is only asserted on compact subsets.

Choose one common regulator subsequence with the following actual classical data.

1. Every fixed baseline covariance entry $C_{0,N}(i,j)$ converges. The baseline has the uniform exponential Schur bounds and positive spectral lower bound used in the strict Gaussian theorem.

2. The strict precision coefficients $B_{N,Y}$, classical base exponents, pivot-log factors, and coherent pre-gauge and joint-chart support coefficients converge locally normally. Both matrix indices of $B_{N,Y}$ belong to Y.

3. All stronger support bounds, analytic neighborhoods, strict sufficient budgets and exact local scalar identities are those of the established finite construction. Aggregate primitive labels by carrier and type before taking limits: there are only finitely many types per fixed carrier, including precision, base, inherited and bad-block types.

These data are chosen independently of the inherited interaction. At fixed parameters their strict families admit a countable diagonal extraction. Each fixed coefficient has finitely many coordinates. For the real fluctuation window, use the actual unwindowed chart's holomorphic extension to a common complex neighborhood and its uniform Cauchy bounds. Bounded measurable functions alone would not give this compactness. The sharp indicators remain fixed; they are not differentiated or subjected to a complex change of variables.

The actual flat formulas already give compatible periodized baseline kernels. More generally, joint entrywise extraction with the stated uniform bounds preserves every finite principal spectral inequality. The conclusions below refer to the selected limiting classical system, not to an arbitrary different sequence of baselines or classical branches.

3.2 Explicit strict transport margins

The common-domain proof permits the following stricter choices. Fix its $B = 53/8$, primitive constants and classical support reserves first. Choose R so that

$$c_1(R)B \le a_0/4, \qquad c_1(R) = \frac{5^d 2^d}{k_R}, \qquad k_R = 1 + \lfloor (R-3)/4 \rfloor. \tag{44}$$

After fixing R, choose J and its logarithmic tail collar so that, with the actual tail-to-margin ratio t,

$$t < 1, \qquad D(J) \log \frac{1}{1-t} \le a_0/4. \tag{45}$$

This is possible because its sufficient bound is a constant times $J^{2d} e^{-\kappa_0 \ell_{\mathrm{tail}}/2}$, which tends to zero with the established choice of collar. It is independent of the later choice of b. Thus

$$\delta_{\mathrm{card}} := a_0 - c_1(R)B - D(J) \log \frac{1}{1-t} \ge a_0/2 > 0. \tag{46}$$

The original weak half allocations alone would only give a nonnegative margin; (44)–(45) make the needed strictness explicit.

In the positive joint core/tail majorant, restricting original carriers to $|H| > Q$ now gives the bound

$$C\epsilon e^{-\delta_{\mathrm{card}} Q}. \tag{47}$$

Choose also a fixed $\zeta > 1$ sufficiently close to one that

$$\zeta t < 1, \qquad c_1(R)B + D(J) \log \frac{1}{1-\zeta t} < a_0. \tag{48}$$

Replacing t by ζt in the exact scalar-multiindex majorant gives an $O(\zeta^{-k})$ bound above tail-expansion degree k. If $t = 0$, there are no positive tail degrees. The rooted spatial weight and the strict spatial budget $\kappa_{\mathrm{tail}} > B + \delta$ control omitted remote supports. The whole output-fiber maximum remains inside the support sum.

Choose b, then the permitted T, g, within the common-domain order with strict slack in the precision, density, bad-block and multiplier inequalities. Equivalently use the smaller quarter budgets of the two-input comparison. Strictness allows a fixed bookkeeping dilation greater than one in the positive majorants below. All these choices are uniform in volume and in preceding M when the established classical estimates are uniform there.

3.3 Continuity of the actual strict Gaussian construction

Lemma 3.1 (Local construction continuity)**.** *Use fixed finite-dimensional fluctuation cells, fixed real windows, the classical data above, and locally normal multiplier coefficients under uniform strict budgets. Fix deterministic connecting graphs that agree on every bounded lifted configuration for sufficiently large periods. The canonical connected Gaussian logarithm coefficients then have unique locally normal limits for those limiting local data and assignments. Their original anchored bound holds on the full open b ball.*

Proof. The proof retains the construction's positive majorants, not merely its final activity norm.

First expand the precision inverse in its B_Y labels. A term has the form

$$C_0(b, u_1) B_{Y_1}(u_1, v_1) C_0(v_1, u_2) \cdots B_{Y_r}(u_r, v_r) C_0(v_r, c).$$

Every intermediate index belongs to a retained precision carrier, because both indices of B_Y lie in Y. The endpoints b, c belong to the finite moment-root set $\mathcal{M}$. If $X = \bigcup_j Y_j$, the final connected carrier contains $\mathcal{M} \cup X$. Masked baseline inverses act on the finite set $\mathcal{M}$. External determinant trace chains have the same property. Consequently a fixed final carrier bounds every primitive spatial label and covariance-chain index. No hidden unbounded index excursion or new spatial reserve is needed. The full C_0 kernel entry is used as an entry, not as an unrecorded walk expansion.

Next keep tags on primitive positive weights and dilate them:

$$B_Y \mapsto zB_Y, \qquad w_Y \mapsto zw_Y, \qquad q_i \mapsto zq_i, \qquad |z| \le 1+\delta_0 \tag{49}$$

for a sufficiently small fixed $\delta_0 > 0$. Here w_Y are interaction multipliers and q_i are bad-block factors. Strict budgets make the same positive overlap, endpoint-tree and Mayer majorants summable on this disc. Scaling a bad-block factor only scales its joint-tail activity weight; no new cutoff or differentiability assertion is involved.

Every nonempty logarithm polymer contains an interaction or bad-block mark. Mayer repetitions repeat such marks. Covariance forest edges number at most the molecule count minus one, while each inverse or determinant density order counts its precision occurrences. Thus the remaining combinatorial tails are $O((1+\delta_0)^{-k})$, together with the strict geometric precision-order tail if counted separately. There is no unmarked unbounded combinatorial order left.

For a fixed final carrier and bounded orders there are only finitely many discrete choices, since primitive labels have already been aggregated by carrier and type. A retained term depends on finitely many baseline entries and analytic precision or multiplier coefficients. Its Gaussian moment uses a finite principal covariance. The partition masks and degree-adapted endpoint perturbations remain in the common accretive neighborhood. Matrix determinants and inverses are continuous holomorphic functions there. The established density majorant is integrable on the entire fixed real Gaussian space, uniformly in the regulator.

Dominated convergence therefore passes the Gaussian integrals and compact forest-parameter integrals. The fixed sharp indicators are only bounded measurable factors; they remain untouched. Multiplier convergence is uniform on their compact retained windows. The argument is uniform on compact background polydiscs, or equivalently follows there from the common holomorphic bounds and uniqueness of the limit.

The order-tail bounds now pass each complete coefficient. Deterministic hull grouping preserves this convergence. Winding configurations cannot remain in a fixed bounded lifted carrier as the period grows. The final anchored cardinality bound controls the omitted large carriers when passing normalized totals on compact fields.

Finally apply Fatou to finite carrier selections at radius $r < b$, then increase r to b and exhaust the carriers. This retains the original full open-domain bound. Compact-total convergence follows by the same cardinality-tail argument as analytic interaction compactness. The result is uniqueness of the local limit for the selected local data and assignments, not convergence in the global anchored norm. □

3.4 Inherited insertions and external factors

Let finite input families converge locally normally with a uniform activity budget. The core/tail construction evaluates each input coefficient at an analytic core in a fixed strict interior of its old b polydisc, then expands only in its small supported tail. It retains the projected input carrier and every attached background or fluctuation support.

Truncate original carrier size, remote support extent and tail-expansion degree. Equations (47)–(48) and the rooted exponential bound give uniform positive tails. On the remaining finite list, ordinary Cauchy estimates pass all needed input derivatives and evaluations. The actual core and fluctuation chart remain in their fixed interior margin, uniformly on the real retained windows. The complete-fiber norm prevents an extra output-coordinate count. Hence Q_{E_N} has locally normal coefficient limits. The compatible assignment is linear in the inherited input. Equal-carrier exponentiation then preserves local convergence and the required multiplier bounds.

Lemma 3.1 applies to $\mathcal{L}_N$. The external pivot factors pass by local analytic composition. The external determinant term passes by its absolutely convergent closed-chain trace series, with every

precision label and its indices retained. Classical substitution passes by the proved summable-field composition limit. We conclude that (43) has locally normal connected coefficient limits under these hypotheses.

3.5 Canonical periodization and quantum quotient descent

The periodization theorem supplies a covariant representative of every translation-invariant class without norm increase, a connected finite-periodic representation at the same a_0, b, and exact independence of that representative as a finite vacuum-normalized total. It also supplies finite local gauge and spatial identities on the common chart. For a fixed covariant representative, its fixed lifted coefficients stabilize exactly by (36); in particular they converge locally normally. On covariant representatives the orbit construction is linear, including on differences.

Theorem 3.2 (Quantum map on the selected local quotient)**.** *Under the fixed classical-system, strict-margin and domain hypotheses above, canonical periodized inputs define a well-defined limiting map $\mathcal{F}_g$ on a sufficiently small ball of invariant analytic interaction classes. Its output has a connected representation at the same a_0, b. The map preserves the local total gauge and spatial identities of the actual sector. Writing $\mathcal{T}$ for classical substitution, the quantum remainder is $\mathcal{G}_g = \mathcal{F}_g - \mathcal{T}$.*

Proof. Choose an allowed covariant representative E. Its finite periodizations and the fixed classical data meet the preceding local-continuity hypotheses. They therefore produce a limiting connected family $\mathcal{F}(E)$.

If E' represents the same class, its finite periodized normalized total is exactly the same. Their actual inherited fluctuation differences agree as scalar insertions, and their background terms agree after the chosen normalization. If raw conventions differ, their constant scalar factor is handled in the separate vacuum normalization. The fixed Gaussian, windows and chart are otherwise identical. Consequently their actual finite output normalized totals agree, with compatible nonvanishing logarithm branches. The canonical coefficient arrays need not agree.

Passing their local limits and then compact normalized totals shows $[\mathcal{F}(E)] = [\mathcal{F}(E')]$. This proves descent. The exact finite local gauge and spatial identities pass on compact backgrounds and their allowed transforms. Holomorphic continuation gives their local complex identities on the common ball. Norm bounds pass by Fatou. The same reasoning applies to $\mathcal{T}$ and hence to the indicated remainder. □

3.6 A quotient Lipschitz estimate without a high-jet hypothesis

Let $\bar{\epsilon}$ be a representative budget for which the strict finite two-input comparison holds. It can be chosen so the base exponent norm v, inherited norms and bad-block budget have the spare allocations used there; for example

$$v \le \eta_*/(64e), \qquad q(E) \le \eta_*/(128e) \quad \text{when } \|E\|_{a_0,b} \le \bar{\epsilon}, \qquad 0 < \eta_* \le 1. \tag{50}$$

The actual linear inherited difference satisfies

$$q(D) \le 16 C_T C_R \frac{gT}{b} \|D\|_{a_0,b}. \tag{51}$$

For two quotient classes of norm at most $\bar{\epsilon}/4$, choose covariant near-minimal representatives Y of y and D of $x - y$, and set $X = Y + D$. For small $\varepsilon > 0$,

$$\begin{aligned} &\|Y\| \le \bar{\epsilon}/4 + \varepsilon, \\ &\|D\| \le \|x - y\|_{\mathcal{Q}} + \varepsilon \le \bar{\epsilon}/2 + \varepsilon, \\ &\|X\| \le 3\bar{\epsilon}/4 + 2\varepsilon < \bar{\epsilon}. \end{aligned} \tag{52}$$

Covariant averaging permits these choices without norm loss. This construction controls the representative difference; two unrelated near-minimal representatives would not do so.

For completeness, interpolate the inherited exponent from Q_Y to $Q_Y + zQ_D$. When $q(D) > 0$, use

$$L_D = \frac{\eta_*}{32e\, q(D)} \geq 2. \tag{53}$$

The strict base, inherited and bad-block budgets keep the Gaussian logarithm family bounded by $S_1\eta_*$ throughout this disc. Banach-valued Cauchy gives the finite difference bound $64eS_1q(D)$. If $q(D) = 0$, the difference is zero. Combining with (51) yields, after local limits, Fatou and $\varepsilon \downarrow 0$,

$$\|\mathcal{G}_g(x) - \mathcal{G}_g(y)\|_{\mathcal{Q}} \leq C_{\text{quant}} \frac{gT}{b} \|x - y\|_{\mathcal{Q}}, \qquad C_{\text{quant}} = 1024eS_1C_TC_R. \tag{54}$$

The pivot and determinant terms cancel because the classical system is input independent. No fifth-order or other jet vanishing of D was used in this logarithm comparison.

If B_R denotes the established bounded classical transport constant at this fixed geometry, then

$$\|\mathcal{F}_g(x) - \mathcal{F}_g(y)\|_{\mathcal{Q}} \leq \left(B_R + C_{\text{quant}} \frac{gT}{b}\right) \|x - y\|_{\mathcal{Q}}. \tag{55}$$

The finite comparison uses the same compatible carrier assignments on both inputs. Its difference family converges locally by the construction just proved, so Fatou applies to that family before the quotient infimum is taken. This is a small inherited-input dependence of the quantum remainder; it is not contraction of the whole unextracted map.

3.7 Varying actual inputs and finite-history compatibility

Theorem 3.3 (Varying-input compatibility)**.** *Let finite input families E_N have exactly invariant normalized totals, uniform strict budgets and a locally normal limit E. Use the same input-independent classical system, fixed windows and exact insertion domains. The actual finite outputs have local limits by the construction above, and their limiting normalized class is $\mathcal{F}_g([E])$. The input and output carrier assignments need not be covariant.*

Proof. Sections 3.3 and 3.4 apply directly to E_N: they require local input convergence and uniform positive tails, not the designation *canonical periodization*. This produces a local output from that particular limiting carrier family.

To identify its class, average each finite E_N over its finite cell-translation group. The norm does not increase. Its raw finite total is unchanged: normalized translation invariance implies raw invariance because the zero field is fixed by translation. Thus the actual finite quantum output normalized total is also unchanged.

Extract a locally normal subsequence of these covariant averages, with limit E^{cov}. Their identical finite normalized totals and interaction compactness imply $[E^{\text{cov}}] = [E]$. The averaged inputs and the canonical periodizations of E^{cov} converge locally normally to the same covariant family. Their input-independent classical data are the same. Construction continuity therefore gives identical canonical limiting output families.

Passing the equality between the original and averaged finite output normalized totals identifies the original limiting class with $\mathcal{F}_g([E])$. This reasoning applies to any extracted covariant average limit, so no interaction-dependent replacement of the classical system is needed. Canonical arrays themselves need not agree. □

By induction, every fixed finite number of actual homogeneous local steps agrees, at the normalized local-total level, with the corresponding composition of limiting quotient maps, provided every intermediate actual finite input remains within the uniform budgets and analytic insertion domains and the classical systems and nested regulators are chosen jointly. At each stage use Theorem 3.3, then the next stage's local continuity. Joint finite lists and countable fixed parameter systems can be extracted by the earlier diagonal construction when their hypotheses hold.

All intermediate windows, Gaussian integrations and generated corrections remain present in this assertion. Each finite vacuum normalization is tracked separately; the quotient statement neither discards it from the integral nor proves convergence of an extensive vacuum energy. This is not an identification of sequentially cut off integrals with one one-shot integral.

An infinite quotient relocalization does not by itself provide small representatives of the actual finite winding remainders. Thus an infinite quotient bound alone does not supply the intermediate finite-input hypotheses just stated. If finite classical reference subtraction is later used, the required interface is local normal convergence of the actual reference families and convergence of its extraction coefficients; equality with a canonical periodized reference is unnecessary and is not asserted here.

3.8 Scope

The result gives an actual local limit and a continuous quantum map for the selected homogeneous local sector and fixed classical system, with the established same-radius analytic continuation and exact insertion representation. It is not an assertion about the full Wilson measure or mixed and large-field histories. It does not prove uniqueness across arbitrary classical subsequences.

The one-input bounds and the small quantum Lipschitz factor do not themselves provide an invariant renormalized ball. A finite block still retains all generated quantum terms; full-reference extraction, amplitude closure, an actual coupling trajectory and the remaining field sectors and observable limits require further work. No infinite trajectory or continuum construction follows merely from the finite-history compatibility proved here.

4 Quadratic forcing from a fixed outer field domain

Fix the composite geometry R, J, carrier exponent $a_0 = 1/16$, and the jointly chosen, input-independent classical system of the preceding local-limit theorem. The gauge Lie algebra is semisimple, in particular it may be $\mathfrak{su}(2)$. Constants may depend on the fixed geometry and color conventions, but are uniform in preceding homogeneous depth M and compatible volume. The input of the quantum step in this section is zero.

4.1 One outer construction and its restrictions

The classical precision, stationary-map, pivot and base-interaction coefficients have independent outer product domains. After R, J are fixed, choose one radius $\rho > 0$ on which all their strict support and Gaussian-neighborhood budgets hold with a fixed margin. No inherited activity is evaluated in this construction. With the actual retained link/color window of threshold T, write

$$\begin{aligned} \mathcal{G}_{M,g,T}(0) &= \Delta_M + \mathcal{Q}_{\mathrm{det},M} + \mathcal{L}_{M,g,T}, \\ \mathcal{L}_{M,g,T}(w) &= \log \mathbb{E}_{C_M(w)}[\chi_T(\xi) e^{V_{\mathrm{base}}(w,\xi;g)}]. \end{aligned} \tag{56}$$

The covariance, stationary branch and charts are independent of g. Field-independent powers of g, flat determinants, Haar constants and the compact gauge-insertion normalization remain in the separately tracked vacuum factor.

The actual outer-domain construction gives an anchored family with

$$\begin{aligned}\|\mathcal{G}_{M,g,T}(0)\|_{a_0,\rho} &\le H(\rho,g,T),\\ H(\rho,g,T) &:= (C_\Delta + C_{\det})\rho + 2S_1\big[e^3\tau(T) + e^{v(g,T)}v(g,T)\big],\\ v(g,T) &\le C_{\text{base}}(gT^3 + gT + g^2T^2),\end{aligned} \tag{57}$$

with harmless fixed enlargements of the constants if needed. Choose $T = T(g) \ge 1$ and $g_0 > 0$ so that every strict base and bad-block budget holds on ρ for $0 < g \le g_0$ and

$$\sup_{M,\,0<g\le g_0} H(\rho,g,T(g)) \le H_* < \infty. \tag{58}$$

A sufficiently large fixed multiple of $\sqrt{\log(1/g)}$, with a fixed lower threshold, is one permitted choice after reducing g_0. This is a bound on a coefficient family, not on an extensive total.

For $0 < b < \rho$, fix these outer data and restrict their coefficients. The zero-input integral on b is exactly the restriction of (56): its chart, covariance, window $T(g)$, measure factors and logarithm branch are unchanged. A field radius enters the estimates, not the defining integral. The infinite statement uses the same outer-domain local-limit system followed by restriction. Unrelated subsequences at different radii are not identified. Two alternative valid carrier assignments for that same finite integral also give the same normalized class by the established local-limit theorem.

4.2 Remove the total constant and linear sectors

Theorem 4.1 (Uniform quadratic zero-input force)**.** *Under (58), for every $b \le \rho/2$ the zero-input correction has a representative of its vacuum-normalized class satisfying*

$$\|[\mathcal{G}_{M,g,T(g)}(0)]\|_{\mathcal{Q},a_0,b} \le C_F b^2, \qquad C_F = \frac{2H_*}{\rho^2}. \tag{59}$$

The constant is independent of b, M and volume. The same representative estimate holds before taking a local limit if the hypotheses are imposed on the actual finite regulators.

Proof. The actual zero-input total is globally adjoint invariant: the real retained-window change of variables, equivalently the local gauge theorem of the common-domain integral, gives this identity. Its derivative at zero in each link slot is an invariant linear functional on the adjoint Lie algebra. It vanishes on commutators, and $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$; hence its total linear jet is zero. This reasoning passes from finite regulators to compact-field derivatives on the chosen limit. Individual carrier coefficients need not have zero linear jets or be separately gauge invariant.

Let P_n be homogeneous phase projection in all background coordinates of one carrier. Cauchy's formula on radii below ρ, followed by their increasing limit, gives

$$\|P_n E_H\|_b \le (b/\rho)^n \|E_H\|_\rho. \tag{60}$$

Replace the outer family by

$$E_H^{\ge 2} := \sum_{n\ge 2} P_n E_H. \tag{61}$$

It retains every declared carrier and represents the same normalized total: constants are normalized away and the sum of the linear sectors is zero. Sum (60) first over degree for each carrier, then over anchored carriers. This yields

$$\|E^{\ge 2}\|_{a_0,b} \le \frac{(b/\rho)^2}{1 - b/\rho}\|E\|_{a_0,\rho} \le \frac{2H_*}{\rho^2}b^2. \tag{62}$$

There is no exchange of a root supremum with an order sum. The argument is identical at finite volume and on the normally convergent infinite family. □

For a Lie algebra with an abelian factor, the adjoint argument alone does not remove a linear total; its vanishing would be an additional hypothesis. Vacuum removal here changes only the normalized class. It does not discard the corresponding physical scalar normalization.

Let the actual full reference satisfy $\beta(S_{MR}) = 1$ and $\|[S_{MR}]\|_{\mathcal{Q},b} \le C_S b^2$, as obtained from its fixed outer classical representation. Then

$$P_{MR} = I - S_{MR}\beta, \qquad \|P_{MR}\| \le p := 1 + C_\beta C_S \tag{63}$$

has a bound independent of b. Therefore

$$\begin{aligned} \|P_{MR}\mathcal{G}_{M,g,T(g)}(0)\|_{\mathcal{Q},b} &\le C_0 b^2, \qquad C_0 = pC_F, \\ |\beta(\mathcal{G}_{M,g,T(g)}(0))| &\le C_\beta C_F. \end{aligned} \tag{64}$$

The b^{-2} factor in the norm of β is present; the proven b^2 forcing cancels it. The finite representative estimate does not by itself define a finite extraction functional: a finite β_N must separately have the corresponding bounded normalization on the actual reference.

4.3 Order of choices and the limits of the conclusion

The outer radius and bound are chosen once after R, J. The final inner radius may subsequently be reduced to meet any fixed finite block of classical transport constraints. For a prescribed family ceiling $\bar{\epsilon}$, the inherited quantum difference coefficient is proportional to $gT(g)/b$. A sufficient budget, with the constants of the common-domain theorem, is

$$\frac{16 C_T C_R g T(g)}{b}\,\bar{\epsilon} \le \frac{\eta_*}{128e}. \tag{65}$$

The denominator $128e$ is the stronger two-input comparison budget used for the quotient Lipschitz estimate. It is met by reducing g_0 after the positive b and $\bar{\epsilon}$ have been fixed. Thus a ceiling of the form $\bar{\epsilon} \le \eta_*/(\text{constant}\, a_Q)$, where $a_Q = 16 C_T C_R g T/b$, creates no radius cycle. The base and bad-block budgets remain separate requirements.

If all intermediate extracted amplitudes have norm at most $C_{\text{int}} b^2$, the bounded classical transport, the force bound and the fixed-coupling quantum Lipschitz bound give an unprojected one-step amplitude bounded by a constant times b^2. Its Maxwell increment is consequently bounded independently of b. This provides neither its sign nor control of infinitely many accumulated increments.

Finally, Δ_M and $\mathcal{Q}_{\det,M}$ have no small factor g. Their removal of constant and linear sectors yields $O(b^2)$, not $O(g^2)$. A bound of the latter kind for the full force would require a separate proved cancellation or subtraction. Nothing here is a contraction in the joint coupling–activity variables, a large-field estimate, or an identification of the local sector with the full Wilson measure.

5 Reference extraction and a contractive finite-block recurrence

Work on the selected joint homogeneous classical system of Sections 1 and 3. Fix the primitive odd ratio ℓ, the elementary composite ratio $R = \ell^s$, the auxiliary grouping J, and carrier exponent $a_0 = 1/16$. At stage j the preceding depth is $M_j = M_0 R^j$. Each stage uses the same cell grouping and analytic radius in its own current units. All constants below may depend on this fixed geometry, but are uniform in M_j. Use a simple gauge Lie algebra, in particular $\mathfrak{su}(2)$, so the Maxwell functional is scalar. The finite family of componentwise references gives the corresponding version for a semisimple product.

The local quantum quotient map is an established input from Theorem 3.2, with its strict domain, periodization, classical-system and logarithm-branch hypotheses. The discussion below does not replace that theorem by an inference from compact-field equality alone. The actual finite-regulator history remains a separate issue stated at the end.

5.1 The actual reference transports exactly

Let $U_M(D)$ be the selected ordinary homogeneous minimum and let

$$S_M(D) = S_{\mathrm{bare}}(U_M(D)), \qquad S_M(1) = 0, \qquad P_{M,R}(W) = \mathcal{M}_\ell^k U_{MR}(W), \quad M = \ell^k. \tag{66}$$

The trace convention and plaquette counting are fixed, and the coupling is outside S_M. Classical background substitution is

$$(\mathcal{T}_{M,R}E)(W) = E(P_{M,R}(W)).$$

It contains no fluctuation integration.

Lemma 5.1 (Exact reference identity)**.** *On the specified common local analytic domains in dimension four,*

$$\mathcal{T}_{M,R}S_M = S_{MR}. \tag{67}$$

This identity holds for the actual ordinary classical reference, not merely for a reference with the same quadratic normalization. It passes to the jointly chosen infinite normalized classes.

Proof. On a finite compatible regulator put

$$U_2 = U_{MR}(W), \qquad D = \mathcal{M}_\ell^k U_2, \qquad U_1 = U_M(D).$$

Both fine configurations are fields on the same oriented graph. Its fine spacing is $1/M$ in D units and $1/(MR)$ in W units. Exact ordinary composition gives

$$\mathcal{M}_\ell^k U_1 = D = \mathcal{M}_\ell^k U_2, \qquad \mathcal{M}_\ell^{k+s} U_1 = W = \mathcal{M}_\ell^{k+s} U_2. \tag{68}$$

The actual minimum comparison balls can be made common for fixed R. Indeed a fine physical potential changes units by $A_1^W(x) = RA_1(Rx)$, so

$$\|A_1^W\|_{C^{1,\alpha}} \le CR^{2+\alpha}\|A_1\|_{C^{1,\alpha}}. \tag{69}$$

The prefix bound is $\|\log D/i\|_\infty \le C_A\|\log W/i\|_\infty/R$. Thus, for fixed R, a sufficiently small real W ball places both fields in both needed comparison classes, uniformly in preceding refinement and volume.

The minimum at D gives $S_{\mathrm{bare}}(U_1) \le S_{\mathrm{bare}}(U_2)$; the minimum at W gives the reverse inequality. In dimension four the coupling-free bare action is the same plaquette sum under this coordinate relabeling: its physical prefactor is $\eta^{d-4} = 1$. Hence the two actions agree. This is also the identity $S_M(V^\star) = S_M(V_p) = S_{MR}$ in the actual common-domain factorization.

Both sides are holomorphic on the specified connected common domain. Uniqueness from the real germ continues the identity there. For $R > C_A$, the angle estimate maps the common b ball into itself; the smaller real comparison ball does not cause a repeated analytic radius loss. No minimum assertion at complex fields is used.

For the fixed joint infinite system, pass vacuum-normalized values on compact W using the connected interaction bounds and summable-field continuity. These are the same hypotheses used for the classical substitution limit. No convergence of an extensive infinite action value is asserted. The finite references vanish at the trivial field, so the normalized identity has no vacuum ambiguity. □

The exact flat midpoint symbol and its identification with the classical Hessian give

$$q_M(p) = |p|^2 I - pp^T + O(|p|^4), \qquad \beta(S_M) = 1. \tag{70}$$

This normalization was proved by the flat symbol calculation; ellipticity alone would not imply it. The fixed-outer-radius classical support estimate and its zero total linear jet also give

$$\|[S_M]\|_{\mathcal{Q},b} \le C_S b^2, \tag{71}$$

uniformly in M and in the subsequent inner radius b. An arbitrary Maxwell-normalized local Wilson reference need not obey (67) as a full nonlinear functional.

5.2 The projection telescope

Write $\mathcal{Q}_j$ for the invariant quotient at stage j and set

$$S_j = [S_{M_j}], \qquad P_j = I - S_j\beta_j, \qquad \mathcal{T}_j = \mathcal{T}_{M_j,R}, \qquad A_j = P_{j+1}\mathcal{T}_j. \tag{72}$$

The notation $S_j\beta_j$ denotes a rank-one operator. Because $\beta_j(S_j) = 1$, P_j projects onto $\ker\beta_j$. The established coefficient and reference estimates give

$$\|\beta_j\| \le C_\beta b^{-2}, \qquad \|P_j\| \le p := 1 + C_\beta C_S. \tag{73}$$

The b^{-2} factor has been cancelled by (71), not omitted.

Lemma 5.2 (Cancellation of every intermediate extraction)**.** *The actual references satisfy*

$$P_{j+1}\mathcal{T}_jP_j = P_{j+1}\mathcal{T}_j. \tag{74}$$

Consequently, for every finite q,

$$A_{j+q-1}\cdots A_j = P_{j+q}\mathcal{T}_{j+q-1}\cdots\mathcal{T}_j, \tag{75}$$

$$A_{j+q-1}\cdots A_jP_j = P_{j+q}\mathcal{T}_{M_j,R^q}P_j. \tag{76}$$

The second identity uses the classical semigroup on the same joint invariant system.

Proof. For $x \in \mathcal{Q}_j$, the difference between the two sides of (74) is $-\beta_j(x)P_{j+1}\mathcal{T}_jS_j$. By (67), this is $-\beta_j(x)P_{j+1}S_{j+1} = 0$. Repeated application removes the intermediate projections. The actual scalar semigroup modulo gauge, together with the universal summable-gauge identity of Section 1, gives (76). □

The proof does not require $\beta_{j+1}\mathcal{T}_j = \beta_j$. Nor does it require equality of sequential and direct carrier arrays. The common scalar class may be estimated using its separately justified direct connected representation. If instead $\mathcal{T}_jS_j = S_{j+1} + D_j$, the error in (74) is $-P_{j+1}D_j\beta_j$, potentially costing $b^{-2}\|D_j\|$. The actual identity makes that error exactly zero.

Use the supported low-kernel and high-jet transport estimates on the same infinite system. Their explicit common-domain and reserve hypotheses give

$$\left\|A_{j+q-1}\cdots A_j|_{\ker\beta_j}\right\| \le \theta_q := p^2\left[C_{\mathrm{low}}(J)R^{-q\alpha} + 6C_{\mathrm{high}}R^{-q}\right]. \tag{77}$$

The fixed grouping J is chosen before q; its finite constant is not assumed uniform as J grows. Choose a finite q so that $\theta_q \le 1/4$. Then choose b to meet the direct R^q high-jet disk and all classical margins. The constants in (77) do not deteriorate just because b is reduced. Bounded one-step transport gives constants

$$\|\mathcal{T}_j\| \le B_T, \qquad \|A_j\| \le B, \qquad B \ge \max(1, pB_T), \tag{78}$$

uniform in j. These may be large; no one-step classical contraction is asserted.

5.3 The local quantum remainder at fixed coupling

Let $\mathcal{F}_j$ be the actual limiting local quotient map with specified coupling g_j, retained threshold $T_{\mathrm{cut},j}$, and the fixed classical system. Write

$$\mathcal{F}_j(x) = \mathcal{T}_jx + \mathcal{G}_j(x), \qquad \Psi_j(x) := P_{j+1}\mathcal{F}_j(x) = A_jx + H_j(x), \qquad H_j = P_{j+1}\mathcal{G}_j. \tag{79}$$

The remainder includes the pivot and Gaussian determinant terms and the full base Gaussian logarithm. Those terms are not discarded. The output of Ψ_j belongs to $\ker \beta_{j+1}$.

Let $\bar{\epsilon}$ be the representative ceiling from the strict two-input comparison. The quotient maps are defined on the ball of radius $\bar{\epsilon}/4$. Equation (54) gives

$$\|\mathcal{G}_j(x) - \mathcal{G}_j(y)\|_{\mathcal{Q}} \le L_{\mathrm{quant},j}\|x - y\|_{\mathcal{Q}}, \qquad L_{\mathrm{quant},j} = C_{\mathrm{quant}}\frac{g_j T_{\mathrm{cut},j}}{b}. \tag{80}$$

It has no equal-low-jet hypothesis. One can see the domain margin directly from (52): choose covariant near-minimal representatives for y and $x - y$ and add them. Both complete inputs then have norm less than $\bar{\epsilon}$. Independent near-minimal representatives for x and y would not control their difference.

Choose a uniform number ν with

$$pL_{\mathrm{quant},j} \le \nu, \qquad \|H_j(x) - H_j(y)\| \le \nu\|x - y\|. \tag{81}$$

All comparisons here hold the coupling, threshold, classical saddle and cutoff data fixed at each step. The limiting map is well defined by Theorem 3.2; the finite comparison and its passage to the quotient are provided by Section 3.6.

There are two useful forcing estimates. The original common-domain bound gives

$$\begin{aligned} &\|H_j(0)\| \le f, \\ &\quad f \le p \sup_j \left[(C_\Delta + C_{\mathrm{det}})b + 2S_1\{e^3\tau(T_{\mathrm{cut},j}) + e^{v_j}v_j\}\right], \\ &\quad v_j \le C_{\mathrm{base}}(g_j T_{\mathrm{cut},j}^3 + g_j T_{\mathrm{cut},j} + g_j^2 T_{\mathrm{cut},j}^2). \end{aligned} \tag{82}$$

More strongly, fix the outer zero-input construction of Section 4 and then restrict to $b \le \rho/2$. Equations (59) and (64) yield

$$\|\mathcal{G}_j(0)\| \le C_F b^2, \qquad \|H_j(0)\| \le C_0 b^2, \qquad C_F = \frac{2H_*}{\rho^2}, \quad C_0 = pC_F. \tag{83}$$

These are the same actual maps restricted from a common outer domain, not radius-dependent replacements. The force need not be $O(g_j^2)$.

5.4 A finite-block perturbation estimate

Lemma 5.3 (Duhamel bound with a contractive classical block)**.** *Suppose the maps in (79) are evaluated on their common admissible neighborhood, (77) and (78) hold, and (81) holds. For two trajectories using the same prescribed step couplings and other fixed data, with their initial difference in* $\ker \beta_j$*, one has*

$$\begin{aligned} \|\delta_q\| &\le \left[\theta_q + \nu\sum_{k=0}^{q-1} B^{q-1-k}(B+\nu)^k\right]\|\delta_0\| \\ &= \left[\theta_q + (B+\nu)^q - B^q\right]\|\delta_0\| \\ &\le \left[\theta_q + q\nu(B+\nu)^{q-1}\right]\|\delta_0\|. \end{aligned} \tag{84}$$

The sum formula includes $\nu = 0$ by continuity. If $\|H_i(0)\| \le f$, then any trajectory obeys

$$\|x_{j+k}\| \le (B+\nu)^k\|x_j\| + f\sum_{l=0}^{k-1}(B+\nu)^l \tag{85}$$

as long as the intermediate maps remain admissible.

Proof. The exact variation-of-constants identity is

$$\begin{aligned}\delta_q &= A_{j+q-1}\cdots A_j\delta_0 \\ &\quad + \sum_{k=0}^{q-1} A_{j+q-1}\cdots A_{j+k+1}\big[H_{j+k}(x_{j+k}) - H_{j+k}(y_{j+k})\big].\end{aligned} \tag{86}$$

An empty product is the identity. The elementary difference bound is $\|\delta_k\| \le (B+\nu)^k\|\delta_0\|$. Apply the improved classical product bound only to the first term in (86); bound each shorter product by the corresponding power of B. This proves the first line of (84). The difference-of-powers identity gives the second line, and bounding every sum term by $(B+\nu)^{q-1}$ gives the third.

For a single step, $\|\Psi_i(x)\| \le (B+\nu)\|x\| + f$. Induction proves (85). No contraction of a shorter classical product was presumed. □

The leading classical term in (84) has no growing factor from intermediate projections. The remaining perturbative sum has finite length q, fixed before the coupling is made small.

5.5 An invariant quadratic amplitude envelope

Theorem 5.4 (Extracted finite-block contraction on the local quotient)**.** *Use the selected joint infinite homogeneous system, the actual reference family, the transport estimates (77)–(78), the quantum maps of Theorem 3.2, and the fixed-outer-radius force of Theorem 4.1. Choose a finite q with $\theta_q \le 1/4$. Put*

$$K = B+1, \qquad L_q = \sum_{l=0}^{q-1} K^l, \qquad K_{\mathrm{amp}} = \max\{1, 2C_0L_q\}, \qquad C_{\mathrm{int}} = K^qK_{\mathrm{amp}} + C_0L_q. \tag{87}$$

Choose the common positive $b \le \rho/2$ to satisfy all direct R^q classical and high-jet domain restrictions and

$$C_{\mathrm{int}}b^2 < \bar{\epsilon}/4. \tag{88}$$

Finally choose the weak-coupling range and thresholds so that the strict two-input budgets, including (65), hold and

$$\nu \le \min\left\{1, \frac{1}{4qK^{q-1}}\right\}. \tag{89}$$

For each initial stage j, every q-step extracted block maps

$$\{x \in \ker\beta_j : \ \|x\| \le r_b\}, \qquad r_b = K_{\mathrm{amp}}b^2, \tag{90}$$

into the corresponding radius-r_b ball at stage $j+q$, with Lipschitz constant at most $1/2$. Every intervening amplitude is at most $C_{\mathrm{int}}b^2 < \bar{\epsilon}/4$.

For a prescribed countable sequence of couplings in this weak range, with its fixed common classical local-limit system, the extracted quotient recursion is therefore defined for all stages starting in (90). At block endpoints two trajectories using that same sequence contract by 2^{-n} after n blocks. These conclusions also bound a density-preserving running trajectory for as long as it stays in the prescribed weak-coupling regime.

Proof. Start in (90). For $0 \le k \le q$, (85), $\nu \le 1$, and (83) give

$$\|x_{j+k}\| \le K^kK_{\mathrm{amp}}b^2 + C_0b^2\sum_{l=0}^{k-1} K^l \le C_{\mathrm{int}}b^2 < \bar{\epsilon}/4. \tag{91}$$

This induction verifies, before the next step is invoked, that each input lies in the common admissible neighborhood. The same holds for the trajectory starting at zero and for any second input in the ball.

Equation (84) and (89) make the block Lipschitz constant at most $1/4 + 1/4 = 1/2$. Its zero-input output has norm at most $C_0 L_q b^2 \le r_b/2$. Therefore its output at any x in (90) has norm at most $\|x\|/2 + r_b/2 \le r_b$. The output is extracted at every step by definition.

All estimates are uniform in the initial homogeneous depth. Repeat the argument block by block. This constructs the stated infinite sequence of quotient classes for fixed prescribed step data; it asserts no convergence of a single nonautonomous trajectory to a fixed point. Applying the same block Lipschitz bound to two trajectories with identical data gives the 2^{-n} difference estimate. □

The parameter order is noncircular. Choose the primitive data, the elementary R and grouping J with the strict Section 3.2 margins, then choose the finite q using (77). The outer radius and H_* in Section 4 are fixed before the final inner radius. Then choose b by (88) and the direct R^q restrictions. Only afterward decrease the coupling ceiling. For example, thresholds equal to a sufficiently large fixed multiple of $\sqrt{\log(1/g)}$, with a fixed lower threshold, have $gT(g)/b \to 0$, $gT(g)^3 \to 0$ and $\tau(T(g)) \to 0$. Thus the comparison and forcing budgets can be met uniformly on $0 < g \le g_0$. In particular

$$\frac{16 C_T C_R g T(g)}{b}\,\bar{\epsilon} \le \frac{\eta_*}{128 e}$$

is imposed with the stronger two-input budget. The constants p, B, θ_q do not grow just because b is reduced.

For comparison, the weaker force (82) also suffices for a fixed-radius envelope. With the same K, L_q, take

$$r = \frac{\bar{\epsilon}}{8K^q}, \qquad f \le \frac{r}{2L_q}. \tag{92}$$

Then intermediate norms are at most $\bar{\epsilon}/8 + r/2 < \bar{\epsilon}/4$, and the same proof preserves the radius-r extracted ball. The quadratic version is stronger because it controls the Maxwell increment independently of the inner radius.

5.6 Density bookkeeping and bounded coupling increments

Along the quadratic envelope the unprojected one-step value obeys

$$\|\mathcal{F}_j(x_j)\| \le \left[B_T C_{\mathrm{int}} + C_F + L_{\mathrm{quant},j} C_{\mathrm{int}}\right] b^2. \tag{93}$$

This follows by separating $\mathcal{T}_j x_j$, $\mathcal{G}_j(0)$ and $\mathcal{G}_j(x_j) - \mathcal{G}_j(0)$. The norm of β consequently gives

$$|\beta_{j+1}(\mathcal{F}_j(x_j))| \le C_\beta \left[B_T C_{\mathrm{int}} + C_F + L_{\mathrm{quant},j} C_{\mathrm{int}}\right]. \tag{94}$$

The factor b^{-2} has again been paid by a proved b^2 amplitude. This is an absolute bound, with no sign.

For the actual density, write $\gamma_j = g_j^{-2}$. The one-step classical factorization is

$$\exp[-\gamma_j S_{j+1} + \mathcal{F}_j(x_j)].$$

Full-reference extraction gives the exact identity

$$\mathcal{F}_j(x_j) = \beta_{j+1}(\mathcal{F}_j(x_j)) S_{j+1} + \Psi_j(x_j), \qquad \gamma_{j+1} = \gamma_j - \beta_{j+1}(\mathcal{F}_j(x_j)). \tag{95}$$

The negative sign is forced by the displayed classical exponent. A freely prescribed weak coupling sequence need not satisfy (95). The theorem bounds the extracted amplitudes uniformly over such sequences as operator estimates; it identifies a continuation of the same density only along a sequence satisfying this extra update.

Two trajectories that separately update their couplings need not use the same maps: their coupling and threshold parameters also differ. Equation (80) controls only the inherited-input dependence at fixed coupling. Thus the 1/2 block estimate is not a contraction in the joint coupling–activity variables.

For a fixed finite number of steps, (94) permits choosing the initial inverse coupling sufficiently large to retain a positive weak-coupling range. It does not control an infinite sum of increments, establish their sign, or prove asymptotic freedom. It does not prove that a physical running trajectory remains in the regime needed by Theorem 5.4.

5.7 What the quotient envelope does not supply

The conclusions concern the invariant infinite quotient of the actual homogeneous local germ and its specified analytic continuation. They hold for every admissible input class on one fixed joint classical system. No uniqueness across arbitrary classical regulator subsequences is asserted.

Theorem 3.3 identifies fixed finite histories only when their actual finite intermediate input families satisfy the uniform budgets and insertion-domain hypotheses. A small quotient norm, or a newly localized infinite representative, does not automatically provide those bounds for the actual finite winding remainders. For a fixed finite q, crude canonical estimates and an additional radius reduction may furnish a finite history, but that construction must be supplied separately. The infinite quotient envelope is not a regulator-uniform finite RG induction.

If extraction is performed before the local limit, it must use the actual finite reference $S_{M,N}$ and a specified finite coefficient functional β_N with the required normalization and convergence. The finite-volume Ward construction supplies a candidate cutoff-jet functional. One must not assume that $S_{M,N}$ equals the canonical periodization of an independently selected infinite reference. The exact classical identity holds for the actual finite references; compatibility of projected finite histories requires the separate coefficient and family statements.

The quotient removes field-independent totals. Physical vacuum factors remain separately tracked, and their extensive limits are not supplied by this theorem. Large-field and mixed histories, global measure consistency, observables and the continuum limit remain outside this local amplitude result.

The intermediate projection obstruction is therefore resolved: the actual classical reference transports exactly, so its extractions telescope. Combined with the proved local quantum quotient map and quadratic force, this yields the finite-block contraction and invariant amplitude envelope above. It does not constitute a completed Yang–Mills construction.

6 The next construction problem

The normalized local quantum map and its finite-history compatibility now have a specified construction. Their domains and positive majorants are uniform in preceding homogeneous classical depth. They provide the setting in which an equivalent connected representation can be used at the next local integration. Field-independent vacuum factors remain separately tracked; no convergence of an extensive vacuum energy is inferred.

The actual inverse coupling obeys

$$\gamma_{j+1} = \gamma_j - \beta_{j+1}(\mathcal{F}_{j,g_j}(E_j)), \qquad \gamma_j = g_j^{-2}.$$

Completing its weak-coupling trajectory is the next essential step. Absolute increment bounds alone do not establish its direction or permit the ultraviolet cutoff to be removed at a fixed interacting endpoint. The generated drift must be evaluated with the actual determinants, cutoff factors and inherited interaction. Its normalization must agree with the reference action used in the exact density identity.

The activity estimates concern the specified homogeneous small-field sector. The full Wilson construction must also incorporate mixed and large-field histories, identify the continued local representation on their required gauge-orbit domains, and obtain uniform estimates with observable sources. Those estimates feed the compactness, rotation, nontriviality and reconstruction obligations in Notebook 01.

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